

ANTONINE UNIVERSITY

Faculty of Engineering

Department of Technology in Computer Science

Cloud Computing Project

Report

Carried out by

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Technology in Computer Science Student

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Create a new Project

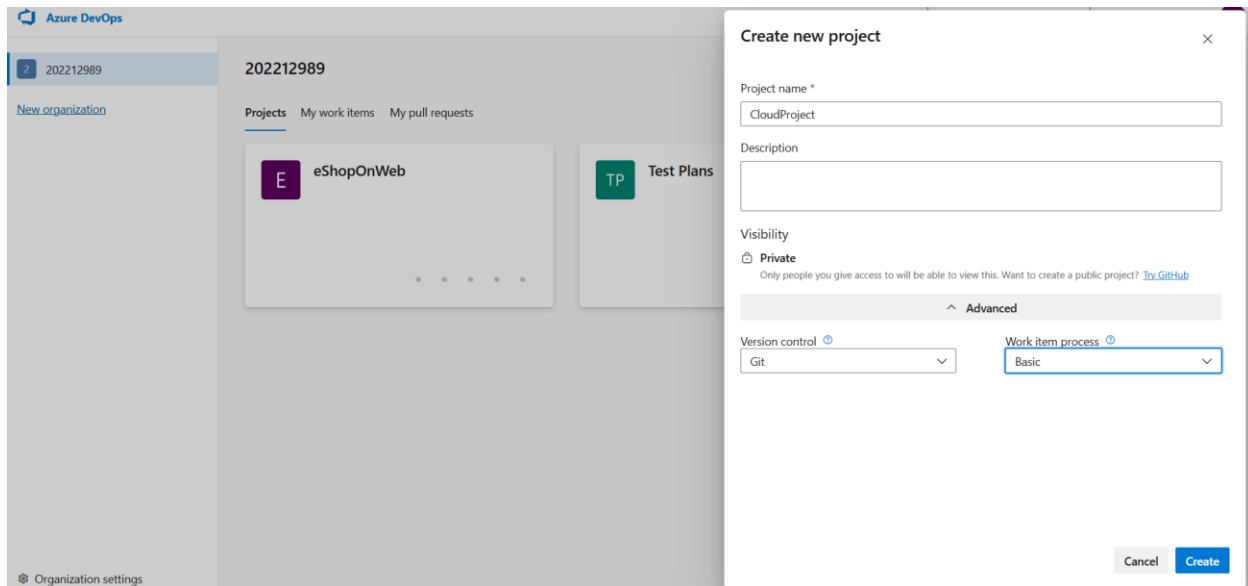


Figure 1_creating azure devops project

Create Checklist on Azure Boards

Open Boards -> Boards and start adding tasks

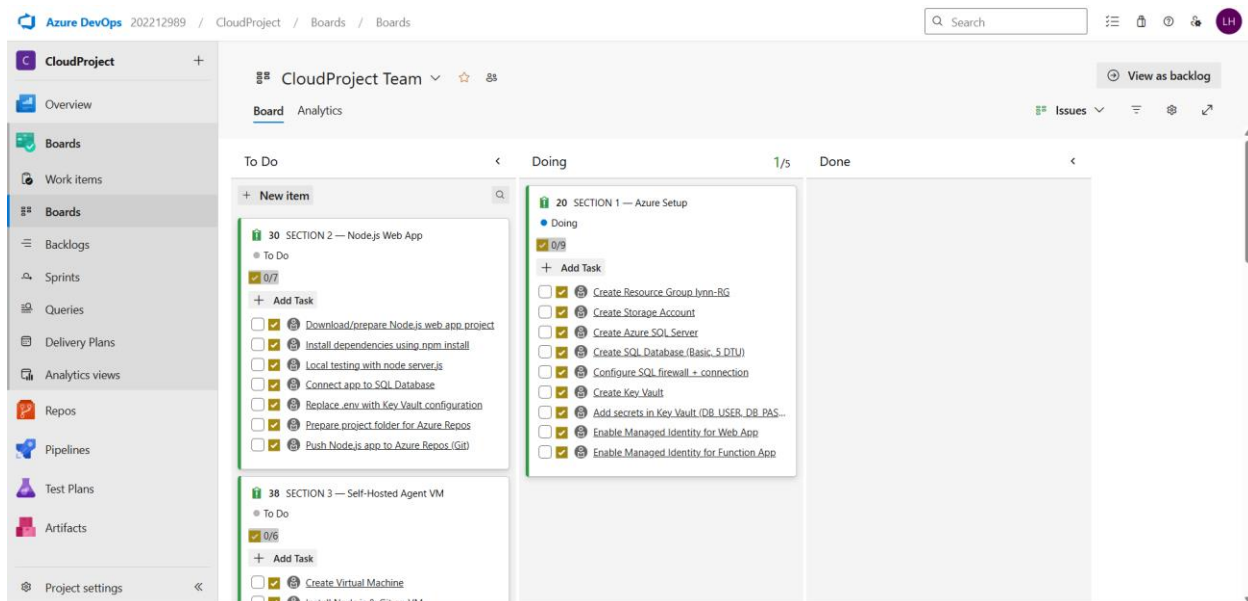


Figure 2_azure boards

 20 SECTION 1 — Azure Setup

● Doing

 0/9

 Add Task

☐





Create Resource Group lynn-RG

☐





Create Storage Account

☐





Create Azure SQL Server

☐





Create SQL Database (Basic, 5 DTU)

☐





Configure SQL firewall + connection

☐





Create Key Vault

☐





Add secrets in Key Vault (DB_USER, DB_PAS...

☐





Enable Managed Identity for Web App

☐





Enable Managed Identity for Function App

Figure 3_azure setup

 30 SECTION 2 — Node.js Web App

● To Do

 0/7

 Add Task

☐





Download/prepare Node.js web app project

☐





Install dependencies using npm install

☐





Local testing with node server.js

☐





Connect app to SQL Database

☐





Replace .env with Key Vault configuration

☐





Prepare project folder for Azure Repos

☐





Push Node.js app to Azure Repos (Git)

Figure 4_node.js webapp

 38 SECTION 3 — Self-Hosted Agent VM

To Do

0/6

+ Add Task

☐

☒



Create Virtual Machine

☐

☒



Install Node.js & Git on VM

☐

☒



Download Azure DevOps Agent

☐

☒



Configure agent & connect it to Agent Pool

☐

☒



Test agent by running sample job

☐

☒



Turn OFF VM when not in use

Figure 5_self hosted agent vm

 45 SECTION 4 — Build Pipeline (Web App CI)

To Do

 Unassigned

Effort

0/5

+ Add Task

☐

☒



Create build pipeline for Node.js app

☐

☒



Add steps: install Node modules, run tests

☐

☒



Publish build artifacts

☐

☒



Enable continuous integration trigger

☐

☒



Run build pipeline using self-hosted agent

Figure 6_build pipeline (CI) webapp

 **71** SECTION 5 — Release Pipeline (Web App CD)

To Do

 0/5

 Add Task

☐





Create release pipeline

☐





Add stage: Deploy to Azure Web App

☐





Connect Key Vault to App Service

☐





Test deployment at Web App URL

☐





Enable continuous deployment trigger

Figure 7_release pipeline cd webapp

 **51** SECTION 6 — Function App Development

To Do

 Lynn HAYDAR

 0/6

 Add Task

☐





Create Azure Function App (Node.js)

☐





Create Azure Functions project locally

☐





Add HTTP Trigger Function

☐





Add Timer Trigger Function

☐





Add Key Vault reference in Function config...

☐





Push Function App code to Azure Repos

Figure 8_function app development

 **58** SECTION 7 — Function App CI/CD

● To Do

 Lynn HAYDAR

 0/6

 Add Task

- ☐   [SECTION 7 — Function App CI/CD](#)
- ☐   [Publish function artifacts](#)
- ☐   [Create release pipeline for Function App](#)
- ☐   [Deploy to Azure Function App](#)
- ☐   [Test HTTP trigger endpoint](#)
- ☐   [Verify timer trigger in Monitor Logs](#)

Figure 9_function app ci/cd

 **65** SECTION 8 — Final Testing & Validation 

● To Do

 Lynn HAYDAR

 0/5

 Add Task

- ☐   [Test Web App homepage](#)
- ☐   [Test /customers SQL endpoint](#)
- ☐   [Test Key Vault connectivity](#)
- ☐   [Test Function App triggers](#)
- ☐   [Validate CI/CD triggers automatically](#)

orkitems/edit/65

Figure 10_final testing + validation

And assign it to yourself

1. Create Azure Resources

1.1. Create Resource Group

[Home](#) > [Resource groups](#) >

Create a resource group ...

Basics

Tags

Review + create

[Automation Link](#)

Basics

Subscription	Lynn Haydar
Resource group name	lynn-RG
Region	France Central

Tags

None

Figure 11_creating resource group

More events in the activity log →

Dismiss all ▼

✓

Resource group created

×

Creating resource group 'lynn-RG' in subscription 'Lynn Haydar' succeeded.

Go to resource group

Pin to dashboard

a few seconds ago

Figure 12_resource group created

1.2. Create Storage Account

Creating a storage account because my function app depends on it and it is used for logs and triggers

Create a storage account

Tables. The cost of your storage account depends on the usage and the options you choose below. [Learn more about Azure storage accounts](#)

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription *	Lynn Haydar
Resource group *	lynn-RG Create new

Instance details

Storage account name * ⓘ	storlynn
Region * ⓘ	(Europe) France Central Deploy to an Azure Extended Zone
Preferred storage type	Choose preferred storage type
Performance * ⓘ	☒ Standard: Recommended for most scenarios (general-purpose v2 account) ☐ Premium: Recommended for scenarios that require low latency.
Redundancy * ⓘ	Locally-redundant storage (LRS)

Figure 13_basics tab storage account

storlynn_1763745801546 | Overview

Deployment

Search

DeleteCancelRedeployDownloadRefresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name: storlynn_1763745801546

Subscription: [Lynn Haydar](#)

Resource group: [lynn-RG](#)

Start time: 21/11/2025, 19:23:25

Correlation ID: 96105241-b833-4eca-9df1-adf6aa09830a

Deployment details

Next steps

Go to resource

Give feedback

Tell us about your experience with deployment

Figure 14_storage account created

1.3. Create Azure SQL Server + SQL Database (5 DTU)

1.3.1. Create SQL Server

[Home](#) > [Azure SQL | SQL databases](#) > [Create SQL Database](#) >

Create SQL Database Server

Microsoft

subscription and resource group as your database.

Server name *

sqlserver-lynn ✓
.database.windows.net

Location *

(Europe) France Central ▼

Authentication

Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#) using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#), or select both SQL and Microsoft Entra authentication.

Authentication method

☐ Use Microsoft Entra-only authentication

☐ Use both SQL and Microsoft Entra authentication

☒ Use SQL authentication

Server admin login *

lynn ✓

Password *

..... ✓

Confirm password *

..... ✓

✓ Password and confirm passv

OK

Figure 15_creating sql server

1.3.2. Configure Compute + Storage

Microsoft Azure

Search resources, services, and docs (G+)

Home > Azure SQL | SQL databases >

Configure

Feedback

Service and compute tier

Select from the available tiers based on the needs of your workload. The vCore model provides a wide range of configuration controls and offers Hyperscale and Serverless to automatically scale your database based on your workload needs. Alternately, the DTU model provides set price/performance packages to choose from for easy configuration. [Learn more](#)

SQL Database Hyperscale: Low price, high scalability, and best feature set. [Learn more](#)

Service tier

Basic (For less demanding workloads)

[Compare service tiers](#)

DTUs

[Compare DTU options](#)

5 (Basic)

Data max size (GB)

2



Cost summary

Basic (Basic)

Cost per DTU (in USD)

DTUs selected

ESTIMATED COST / MONTH

1.22

x 5

6.11 USD

Figure 16_configure database

1.3.3. Create SQL Database

Microsoft Azure

Search resources, services, and i

[Home](#) > [Azure SQL | SQL databases](#) >

Create SQL Database

Microsoft

Subscription * ⓘ

Lynn Haydar

Resource group * ⓘ

lynn-RG

[Create new](#)

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name *

mydb-lynn

Server * ⓘ

(new) sqlserver-lynn (France Central)

[Create new](#)

Want to use SQL elastic pool? ⓘ

☐ Yes ☒ No

Workload environment

☐ Development ☒ Production

ⓘ Default settings provided for Production workloads. Configurations can be modified as needed.

Compute + storage * ⓘ

Basic

2 GB storage

[Configure database](#)

Backup storage redundancy

Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.

Review + create

Next : Networking >

Figure 17_basics tab sql

Basics **Networking** Security Additional settings Tags Review + create

Configure network access and connectivity for your server. The configuration selected below will apply to the selected server 'sqlserver-lynn' and all databases it manages. [Learn more](#)

Network connectivity

Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. [Learn more](#)

Connectivity method * ⓘ

- ☐ No access
- ☒ Public endpoint
- ☐ Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#)

Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server *

No Yes

Add current client IP address *

No Yes

Figure 18_networking tab sql

Create SQL Database ...

Microsoft

Basics Networking Security Additional settings Tags **Review + create**

Product details

SQL database
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Estimated cost per month
6.11 USD

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#).

Basics

Subscription	Lynn Haydar
Resource group	lynn-RG
Region	France Central
Database name	mydb-lynn
Server	(new) sqlserver-lynn
Authentication method	SQL authentication
Server admin login	lynn
Compute + storage	Basic: 2 GB storage
Backup storage redundancy	Geo-redundant backup storage

Networking

Allow Azure services and resources to
access this server. Yes

Create

< Previous

[Download a template for automation](#)

Figure 19_review + create tab sql

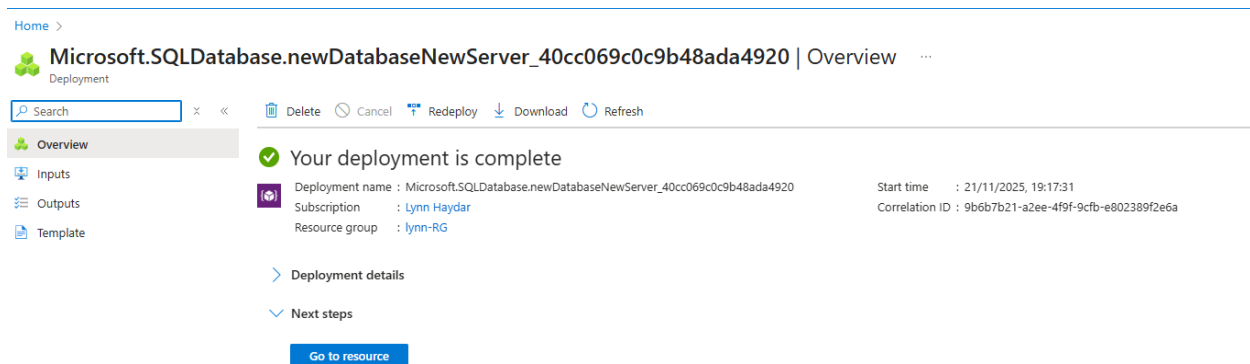


Figure 20_sql db created successfully

1.3.4. Create the Customers Table

My app needs a customers table, I will create it inside azure SQL

Go to SQL Database created -> Query editor (left side) -> Log in using sql admin credentials

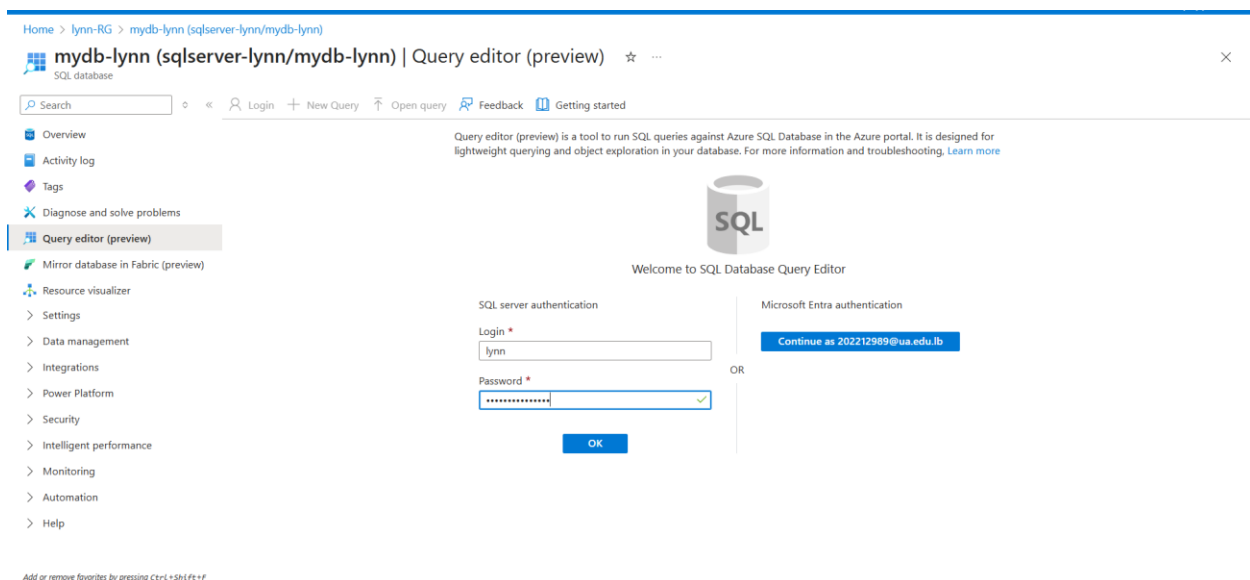


Figure 21_logging in sql

Write this sql query and click run

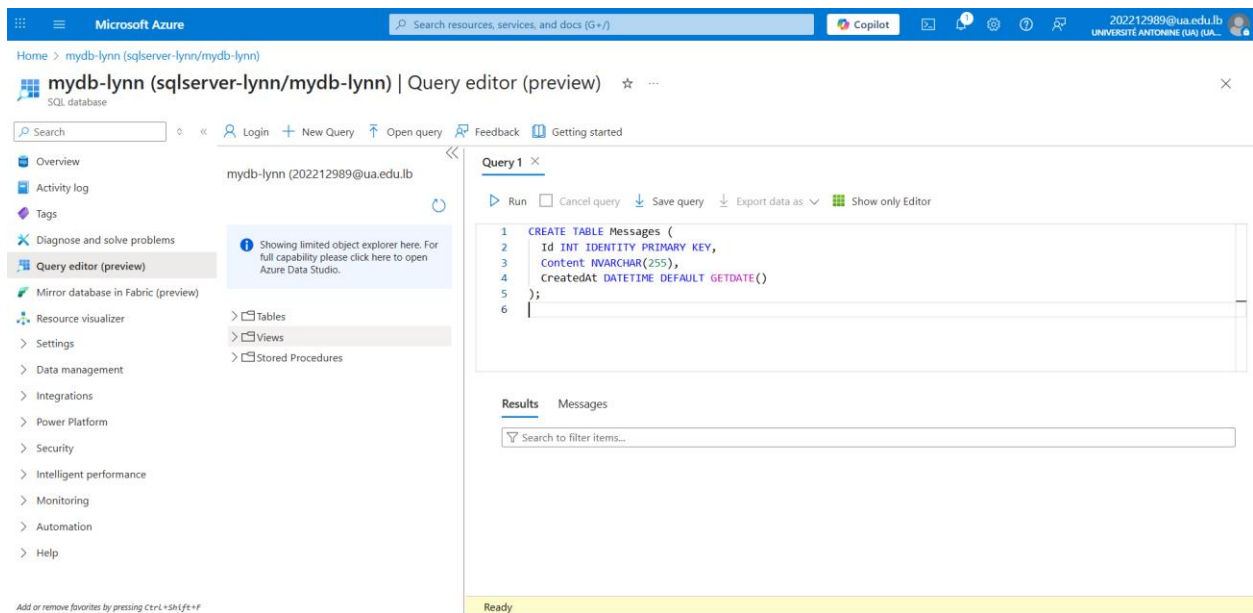


Figure 22_creating table

1.4. Create the Key Vault + Add Secrets

My Node.js Web App and Function App should not store passwords or connection strings in code.

Instead, all secrets will live in **Azure Key Vault**, and both apps will access them through **Managed Identity**.

Create a key vault ...

- Basics
- Access configuration
- Networking
- Tags
- Review + create

Azure Key Vault is a cloud service used to manage keys, secrets, and certificates. Key Vault eliminates the need for developers to store security information in their code. It allows you to centralize the storage of your application secrets which greatly reduces the chances that secrets may be leaked. Key Vault also allows you to securely store secrets and keys backed by Hardware Security Modules or HSMs. The HSMs used are Federal Information Processing Standards (FIPS) 140-2 Level 2 validated. In addition, key vault provides logs of all access and usage attempts of your secrets so you have a complete audit trail for compliance.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Lynn Haydar

Resource group *

lynn-RG

Create new

Instance details

Key vault name * ⓘ

kv-lynn

Region *

France Central

Pricing tier * ⓘ

Standard

Recovery options

Figure 23_basics tab key vault

Delete

Cancel

Redeploy

Download

Refresh

Your deployment is complete

Deployment name : kv-lynn

Subscription : [Lynn Haydar](#)

Resource group : [lynn-RG](#)

Start

Corri

>

Deployment details

>

Next steps

Go to resource

Give feedback

Tell us about your experience with deployment

Figure 24_key vault created

Go to Key vault -> Secrets (objects left side menu) -> Generate/Import

Secret 1: deb-server which is sql server name

Home > [lynn-RG](#) > [kv-lynn](#) | Secrets >

Create a secret

...

Upload options

Manual

Name *

db-server

Secret value *

.....

Content type (optional)

Set activation date

Set expiration date

Enabled

Yes

No

Tags

0 tags

Figure 25_db-server

[Home](#) > [lynn-RG](#) > [kv-lynn | Secrets](#) >

Create a secret ...

Upload options	Manual
Name * ⓘ	db-user
Secret value * ⓘ	...
Content type (optional)	
Set activation date ⓘ	☐
Set expiration date ⓘ	☐
Enabled	Yes No
Tags	0 tags

Figure 26_db-user

[Home](#) > [lynn-RG](#) > [kv-lynn | Secrets](#) >

Create a secret ...

Upload options	Manual
Name * ⓘ	db-password
Secret value * ⓘ	
Content type (optional)	
Set activation date ⓘ	☐
Set expiration date ⓘ	☐
Enabled	Yes No
Tags	0 tags

Figure 27_db-password

Home > lynn-RG > kv-lynn | Secrets >

Create a secret

Upload options: Manual

Name: db-name

Secret value: *****

Content type (optional):

Set activation date: ☐

Set expiration date: ☐

Enabled: Yes No

Tags: 0 tags

Figure 28_db-name

kv-lynn | Secrets

Search

+ Generate/import Refresh Restore Backup Manage deleted secrets View sample code

The secret 'db-name' has been successfully created.

Name	Type	Status	Expiration date
db-name		✓ Enabled	
db-password		✓ Enabled	
db-user		✓ Enabled	
db-server		✓ Enabled	

Figure 29_secrets created

1.5. Create the Azure Web App

This is where my Node.js application will run.

1.5.1. Creating Webapp

[Home](#) > [Create a resource](#) >

Create Web App ...

Project Details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

Lynn Haydar



Resource Group * ⓘ

lynn-RG

[Create new](#)

Instance Details

Name

web-lynn

-avh0g7fxendagjbf.francecentral-01.azurewebsites.net

☒ Secure unique default hostname on. [More about this update](#)

Publish *

☒ Code ☐ Container

Runtime stack *

Node 20 LTS

⚠ Node 20 LTS will reach EOL on 4/30/2026 [Learn More](#)

Operating System *

☒ Linux ☐ Windows

Region *

France Central

i Not finding your App Service Plan? Try a different region or select your App Service Environment.

[Review + create](#)

[< Previous](#)

[Next : Database >](#)

Figure 30_creating web app

Home >

 **Microsoft.Web-WebApp-Portal-9ddd8339-98db** | Overview ⚙ ...

Deployment

Search × <<

Delete Cancel Redeploy Download Refresh

 Overview

 Inputs

 Outputs

 Template

✓ **Your deployment is complete**

 Deployment name: Microsoft.Web-WebApp-Portal-9ddd8339-98db Start time: 23/11/2025, 15:32:18
Subscription: [Lynn Haydar](#) Correlation ID: 4e35947c-f7bb-471a-b7e0-f80c819cfc64 
Resource group: [lynn-RG](#)

▼ Deployment details

^ Next steps

[Manage deployments for your app.](#) Recommended

[Protect your app with authentication.](#) Recommended

[Go to resource](#)

Give feedback

 Tell us about your experience with deployment

Figure 31_webapp created

1.5.2. Enabling managed identity

This allows my app to access Key Vault securely.

Home > Microsoft.Web-WebApp-Portal-9ddd8339-98db | Overview > web-lynn

web-lynn | Identity ☆ ...
Web App

Search

Settings

- Environment variables
- Configuration (preview)
- Instances
- Authentication
- Identity**
- Backups
- Custom domains
- Certificates
- Networking
- WebJobs
- Service Connector
- Locks
- Performance
- App Service plan
- Development Tools
- API
- Monitoring

Add or remove favorites by pressing Ctrl+Shift+F

System assigned User assigned

A system assigned managed identity is restricted to one per resource and is tied to the lifecycle of this resource. You can RBAC). The managed identity is authenticated with Microsoft Entra ID, so you don't have to store any credentials in code

Save Discard Refresh Troubleshoot Got feedback?

Status ⓘ

Off On

Figure 32_enabling identity

1.5.3. Give Web App Access to Key Vault

Go to my key vault -> IAM -> add role assignment -> choose key vault secrets user -> select web-lynn

Microsoft Azure

Home > Resource Manager | Resource groups > lynn-RG > kv-lynn | Access control (IAM) >

Add role assignment

Role **Members** Conditions Review + assign

Selected role Key Vault Secrets User

Assign access to ☐ User, group, or service principal ☒ Managed identity

Members [+ Select members](#)

Name	Object ID	Type
No members selected		

Description Optional

[Review + assign](#) [Previous](#) [Next](#)

Select managed identities

Some results might be hidden due to your ABAC condition.

Subscription * Lynn Haydar

Managed identity App Service (1)

[Search by name](#)

Selected members:

-  web-lynn /subscriptions/3a632cd6-7ed6-4b34-97db-251c7d4cc357/resourceGroups... [Remove](#)

[Select](#) [Close](#) [Feedback](#)

Figure 33_adding role to webapp

Now my web app can read secrets from key vault

1.6. Create Virtual Machine

1.6.1. Creating VM

Microsoft Azure

Search resources, services, and docs (G+/J)

[Home](#) > [Compute infrastructure | Virtual machines](#) > [Create a virtual machine](#) >

Create a virtual machine

Help me choose the right VM size for my workload

Help me create a VM optimized for high availability

Help me create a low cost VM

Help me create a low cost VM

Help me create a VM optimized for high availability

Help me choose the right VM size for my workload

Resource group *

lynn-RG

Create new

Instance details

Virtual machine name *

lynn-agent-vm

Region *

(Europe) France Central

Deploy to an Azure Extended Zone

Availability options

Availability zone

Zone options

☐ Self-selected zone

Choose up to 3 availability zones, one VM per zone

☒ Azure-selected zone (Preview)

Let Azure assign the best zone for your needs

Availability zone

Azure-selected zone (Preview)

Security type

Trusted launch virtual machines

Configure security features

Image *

Windows Server 2022 Datacenter - x64 Gen2

See all images | Configure VM generation

VM architecture

☐ Arm64

☒ x64

Arm64 is not supported with the selected image.

Run with Azure Spot discount

☐

Size *

Standard_B2as_v2 - 2 vcpus, 8 GiB memory (US\$68.77/month)

See all sizes

< Previous

Next : Disks >

Review + create

Figure 34_basics tab vm

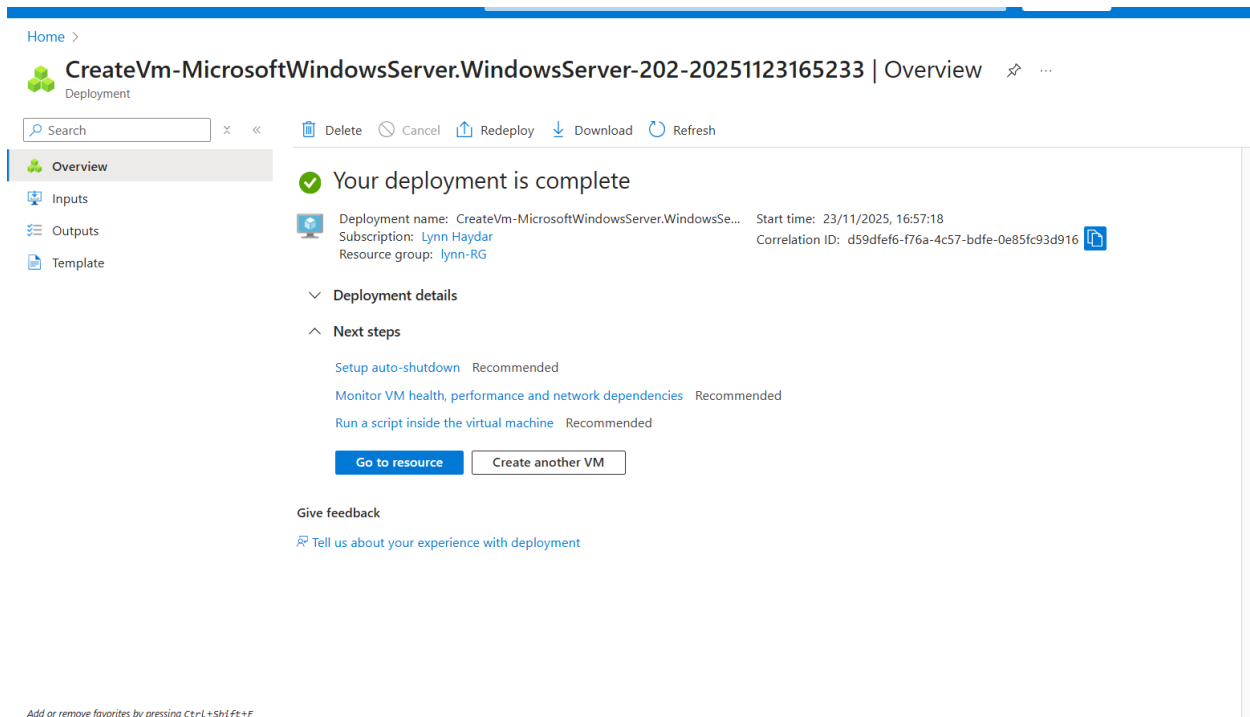


Figure 35_vm created

Go to resource -> connect -> download RDP file and log in

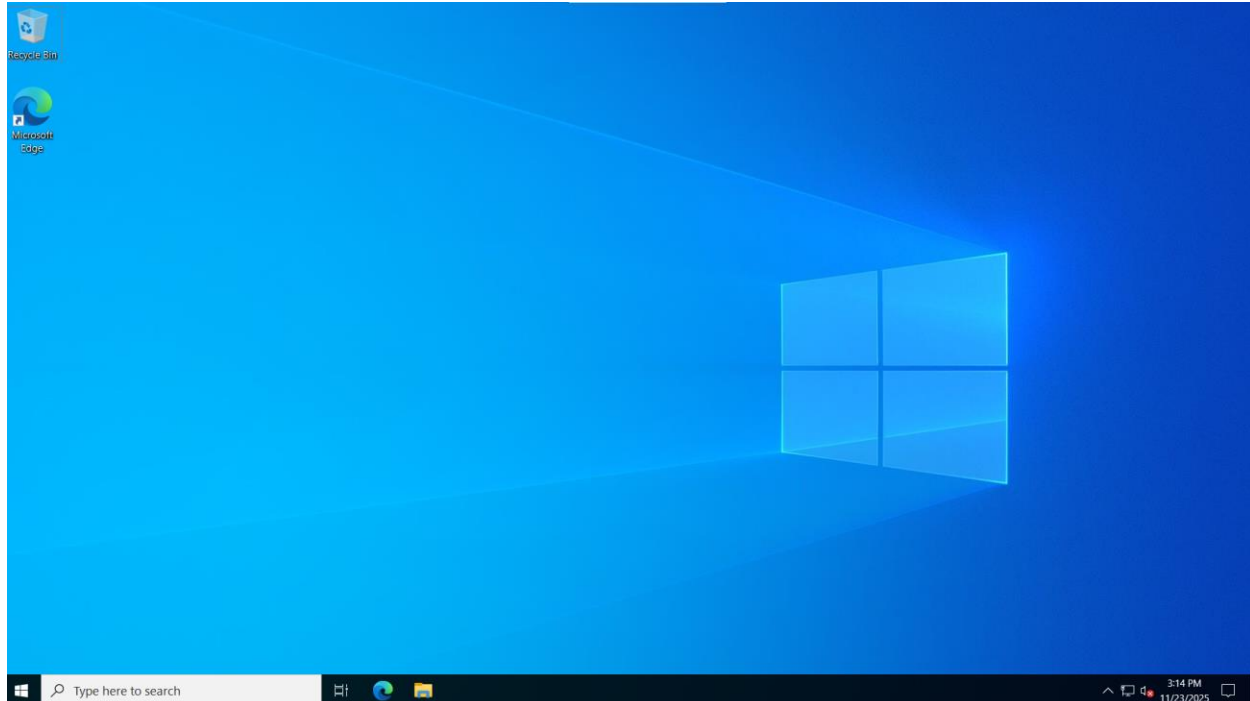


Figure 36_logging in

Install requirements inside the VM: Node.js 20 LTS and GIT

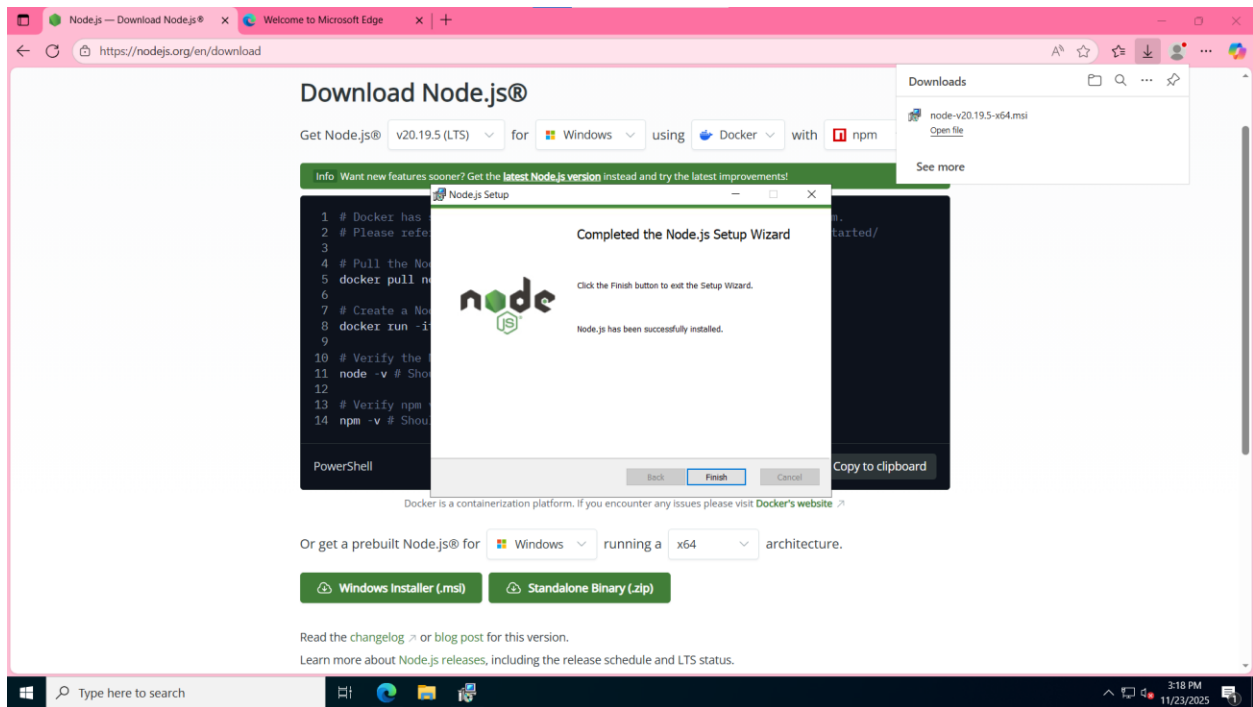


Figure 37_installing node

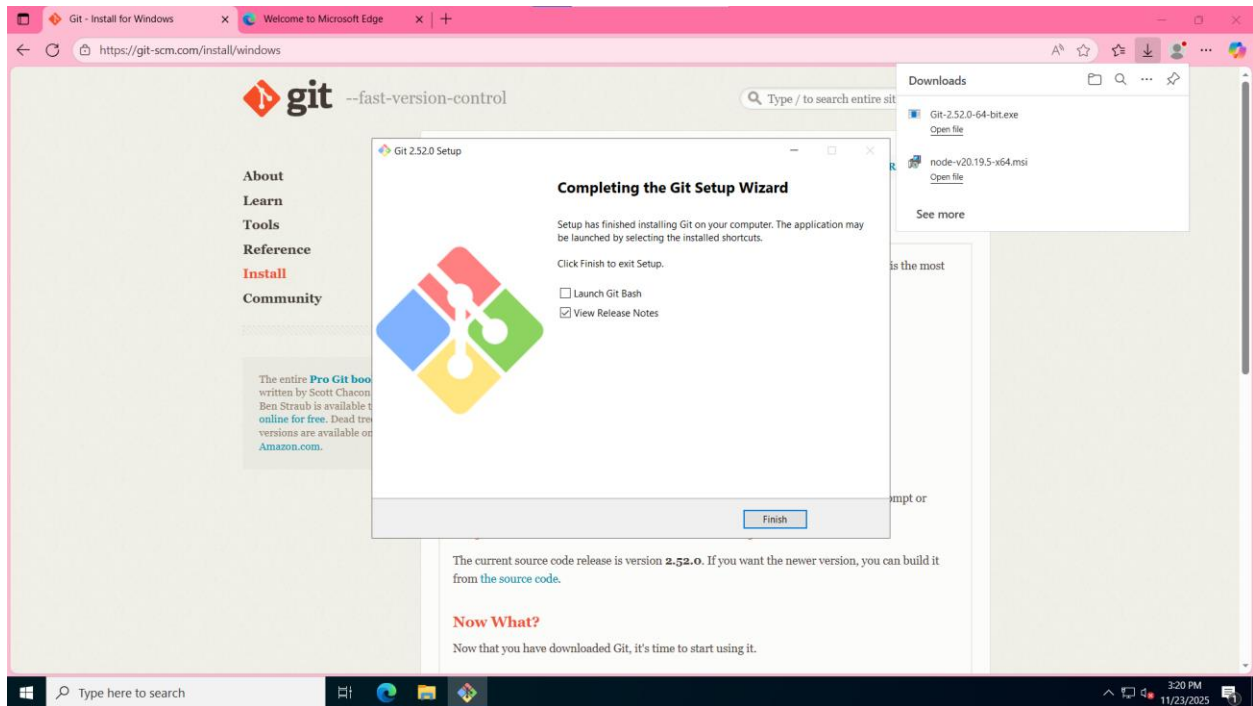


Figure 38_installing git

```
C:\Users\lynn>node --version
v20.19.5

C:\Users\lynn>git --version
git version 2.52.0.windows.1

C:\Users\lynn>_
```

Figure 39_verifying installation

1.6.2. Install agent on VM

Log in to Azure DevOps -> agent pools -> default -> new agent -> download it -> extract -> configure -> run

```
Administrator: C:\Windows\System32\cmd.exe - .\run.cmd
Microsoft Windows [Version 10.0.20348.4405]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lynn\Desktop\agent>config.cmd

Azure Pipelines
agent v4.264.2 (commit 491eca9)

>> Connect:

Enter server URL > https://dev.azure.com/202212989
Enter authentication type (press enter for PAT) >
Enter personal access token > *****
Connecting to server ...

>> Register Agent:

Enter agent pool (press enter for default) >
Enter agent name (press enter for lynn-agent-vm) >
Scanning for tool capabilities.
Connecting to the server.
Successfully added the agent
Testing agent connection.
Enter work folder (press enter for _work) >
2025-11-23 15:31:26Z: Settings Saved.
Enter run agent as service? (Y/N) (press enter for N) >
Enter configure autologon and run agent on startup? (Y/N) (press enter for N) >

C:\Users\lynn\Desktop\agent>.\run.cmd
Scanning for tool capabilities.
Connecting to the server.
2025-11-23 15:32:10Z: Listening for Jobs
```

Figure 40_configure and run agent on vm

Default

Update all agents

New agent

Jobs

Agents

Details

Security

Approvals and checks

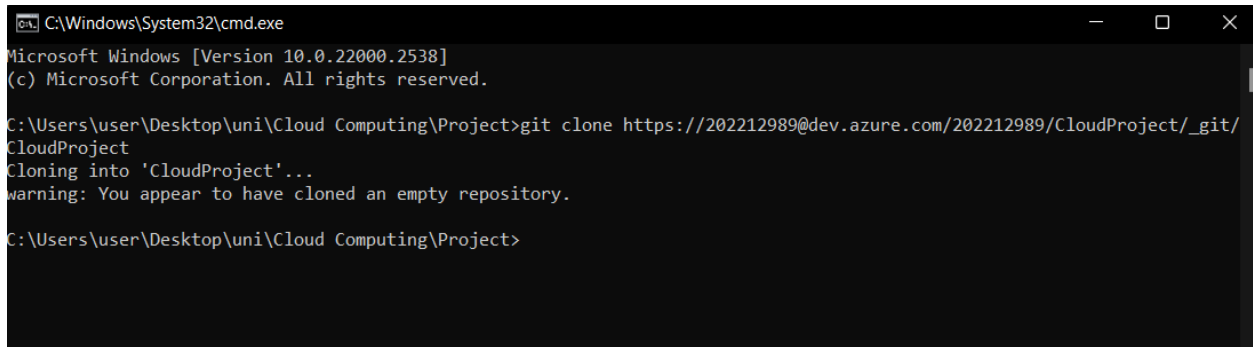
Analytics

Name	Last run	Current status	Agent version	Enabled
lynn-agent-vm Online		Idle	4.264.2	On

Figure 41_agent online

2. Push Node.js WebApp to Azure Repos

Go to Azure DevOps -> Cloud Project -> Repos -> copy link -> clone project locally



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user\Desktop\uni\Cloud Computing\Project>git clone https://202212989@dev.azure.com/202212989/CloudProject/_git/CloudProject
Cloning into 'CloudProject'...
warning: You appear to have cloned an empty repository.

C:\Users\user\Desktop\uni\Cloud Computing\Project>
```

Figure 42_cloning project locally

Copy and paste the project I want to deploy inside that folder

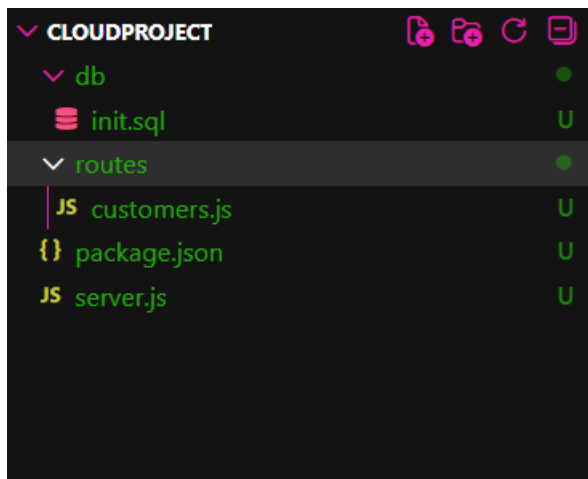
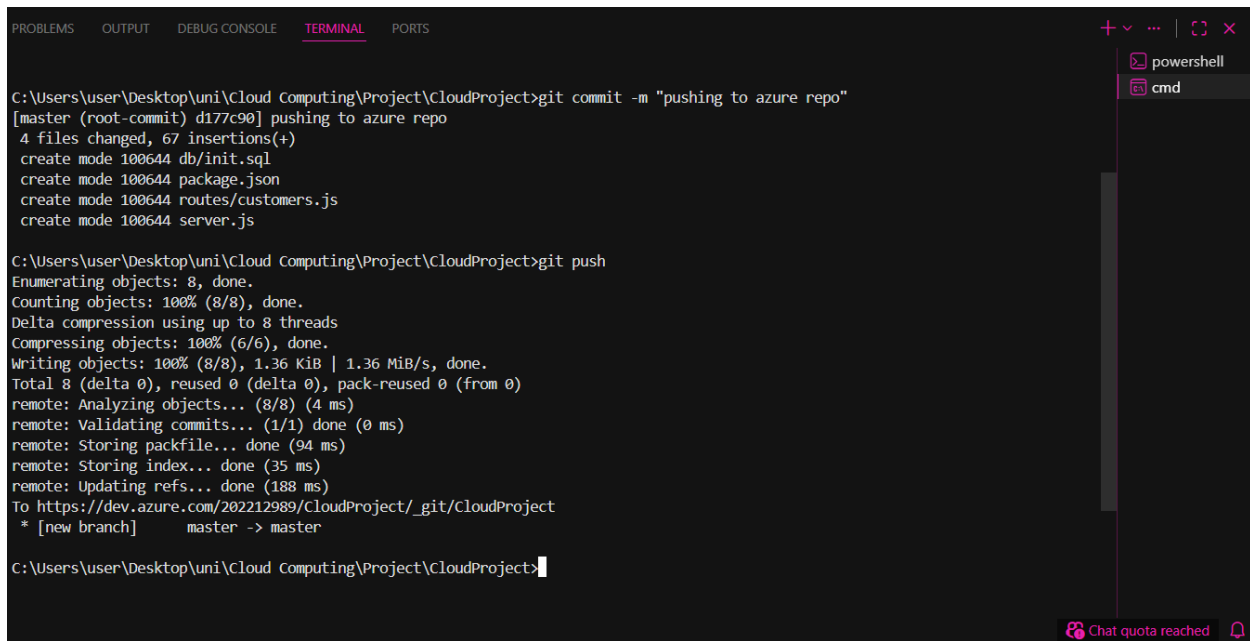


Figure 43_project directory



A terminal window showing the execution of git commands. The first command is `git commit -m "pushing to azure repo"`, which creates four new files: `db/init.sql`, `package.json`, `routes/customers.js`, and `server.js`. The second command is `git push`, which enumerates, counts, compresses, and writes the objects, then pushes them to the remote repository. The output shows the push was successful, with the master branch updated.

```
C:\Users\user\Desktop\uni\Cloud Computing\Project\CloudProject>git commit -m "pushing to azure repo"
[master (root-commit) d177c90] pushing to azure repo
4 files changed, 67 insertions(+)
create mode 100644 db/init.sql
create mode 100644 package.json
create mode 100644 routes/customers.js
create mode 100644 server.js

C:\Users\user\Desktop\uni\Cloud Computing\Project\CloudProject>git push
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (8/8), 1.36 KiB | 1.36 MiB/s, done.
Total 8 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Analyzing objects... (8/8) (4 ms)
remote: Validating commits... (1/1) done (0 ms)
remote: Storing packfile... done (94 ms)
remote: Storing index... done (35 ms)
remote: Updating refs... done (188 ms)
To https://dev.azure.com/202212989/CloudProject/_git/CloudProject
 * [new branch]      master -> master

C:\Users\user\Desktop\uni\Cloud Computing\Project\CloudProject>
```

Figure 44_pushing to azure repos

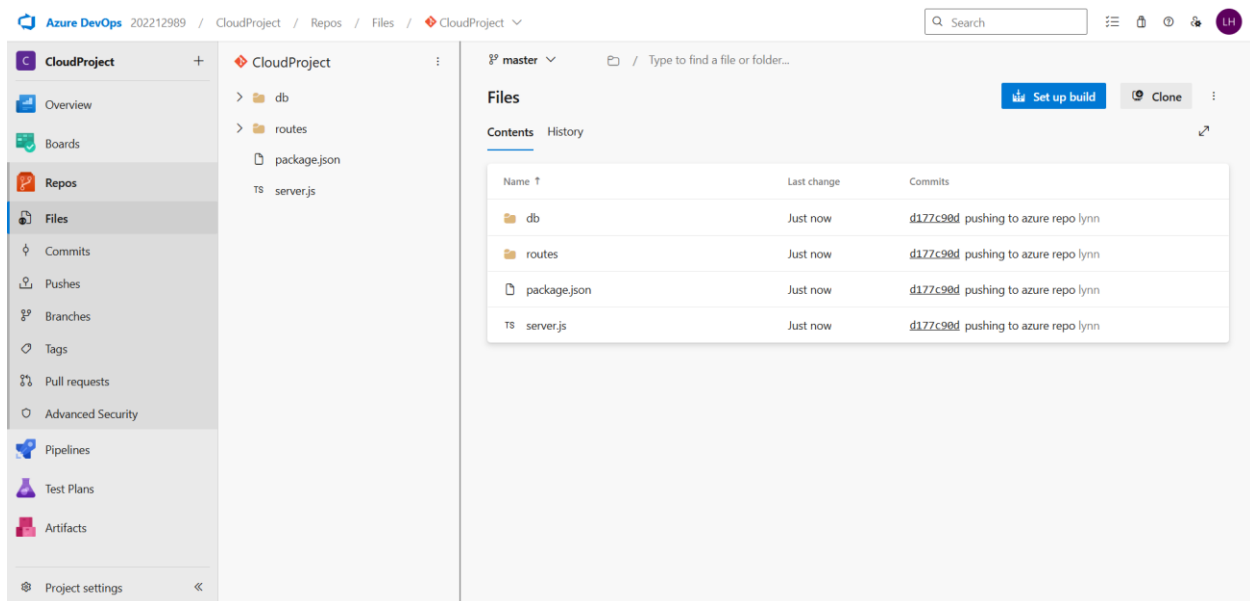


Figure 45_push successful

3. Create the build Pipeline (CI)

Go to Pipelines -> Create Pipelines -> Azure repos git -> choose my project -> starter yaml (modify it) -> save and run

← CloudProject

```
master ▾ CloudProject / azure-pipelines.yml

1 trigger:
2   - main
3
4 pool:
5   - name: 'Default'
6   - demands:
7     - agent.name -equals lynn-agent-vm
8
9 steps:
10  Settings
11  - task: NodeTool@0
12    inputs:
13      - versionSpec: '20.x'
14      - displayName: 'Install Node.js'
15  - script: npm install
16    displayName: 'Install dependencies'
17  - script: echo "Build step"
18    displayName: 'Build'
19
20  Settings
21  - task: ArchiveFiles@2
22    inputs:
23      - rootFolderOrFile: '$(System.DefaultWorkingDirectory)'
24      - includeRootFolder: false
25      - archiveType: 'zip'
26      - archiveFile: '$(Build.ArtifactStagingDirectory)/nodeapp.zip'
27      - replaceExistingArchive: true
28
29  Settings
30  - task: PublishBuildArtifacts@1
31    inputs:
32      - PathToPublish: '$(Build.ArtifactStagingDirectory)'
33      - ArtifactName: 'drop'
34      - publishLocation: 'Container'
```

Figure 46_yaml pipeline

this pipeline:

- Installs Node.js 20
- Installs dependencies (npm install)
- Creates a ZIP file containing entire project
- Publishes the ZIP as a build artifact (for the Release pipeline)

All pipelines > New release pipeline

Pipeline Tasks Variables Retention Options History

Deploy to WebApp
Deployment process

Run on agent
Run on agent

Deploy Azure App Service
Azure App Service deploy

Display name *
Deploy Azure App Service

Connection type * ⓘ
Azure Resource Manager

Azure subscription * ⓘ | Manage
Lynn Haydar (3a632cd6-7ed6-4b34-97db-251c7d4cc357)
ⓘ Scoped to subscription 'Lynn Haydar'

App Service type * ⓘ
Web App on Linux

App Service name * ⓘ
web-lynn

☐ Deploy to Slot or App Service Environment ⓘ

Package or folder * ⓘ
\$(System.DefaultWorkingDirectory)/**/*.zip

Runtime Stack ⓘ

Figure 49_configuring agent

Now add the artifact

×

Add an artifact

Source type


✓ Build


Azure Repos ...


GitHub


TFVC

[5 more artifact types](#) ▼

Project * ⓘ

CloudProject ▼

Source (build pipeline) * ⓘ

CloudProject ▼

Default version * ⓘ

Latest ▼

Source alias * ⓘ

_CloudProject

ⓘ The artifacts published by each version will be available for deployment in release pipelines. The latest successful build of **CloudProject** published the following artifacts: **drop**.

Add

Figure 50_adding artifact

Enable Continuous Deployment Trigger

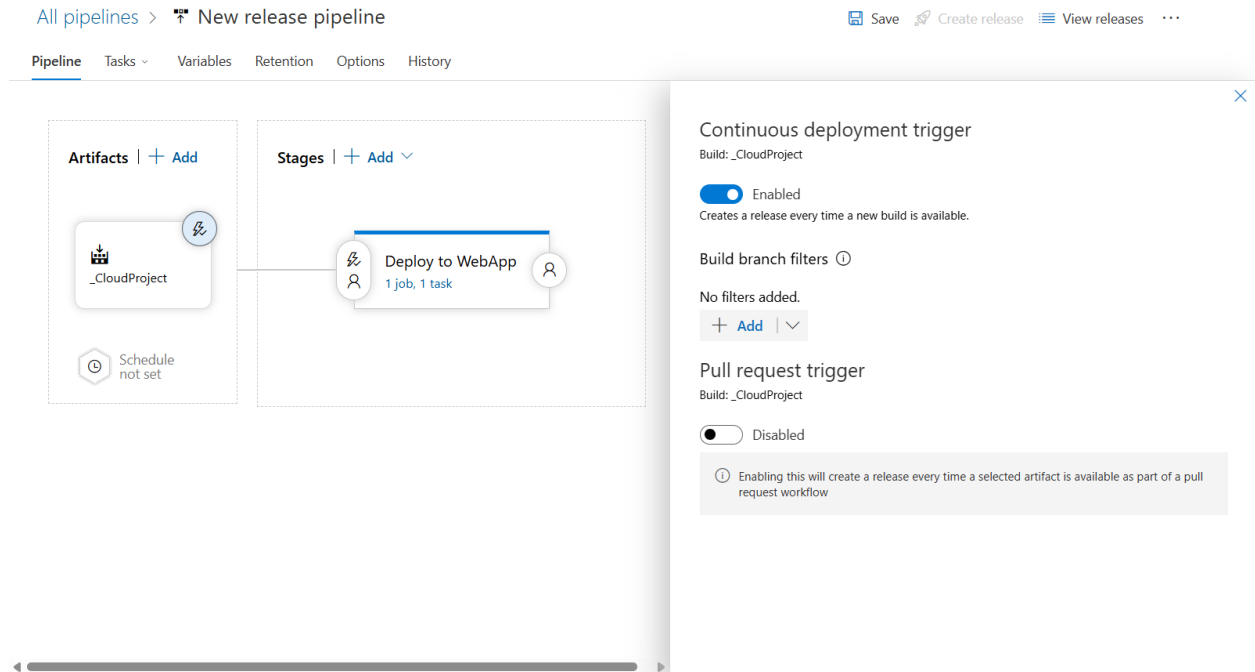


Figure 51_enabling continuous deployment

This means every time I push code → CI → CD → Web App deploy.

Now Create Release

Create a new release



New release pipeline

Pipeline ^

Click on a stage to change its trigger from automated to manual.

Deploy to Wel

Stages for a trigger change from automated to manual.

Artifacts ^

Select the version for the artifact sources for this release

Source alias	Version	
_CloudProject	20251123.4	

Release description

CreateCancel

Figure 52_creating release

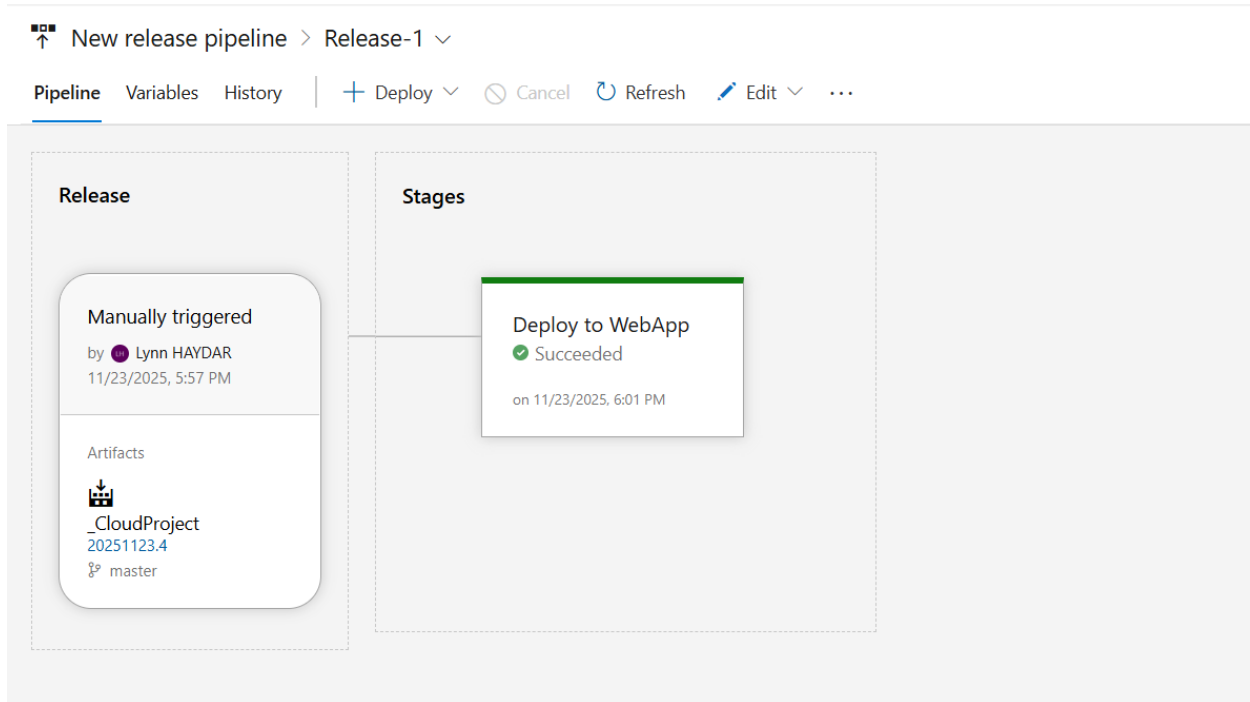


Figure 53_deploy succeeded

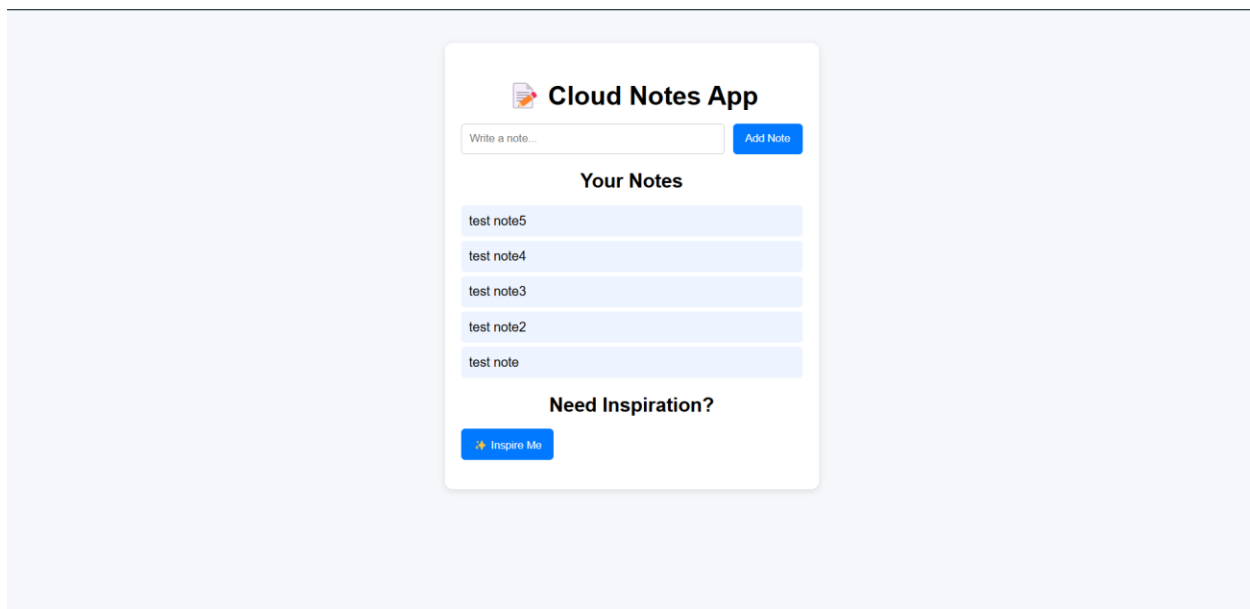


Figure 54_testing result

5. Configure Web App to use Key Vault Secrets

Right now, my app deploys, but it cannot connect to SQL yet

Previously we turned managed identity on and added role assignment.

Now go to environment variables -> app settings and add the variables, copy the link from key vault kv-lynn

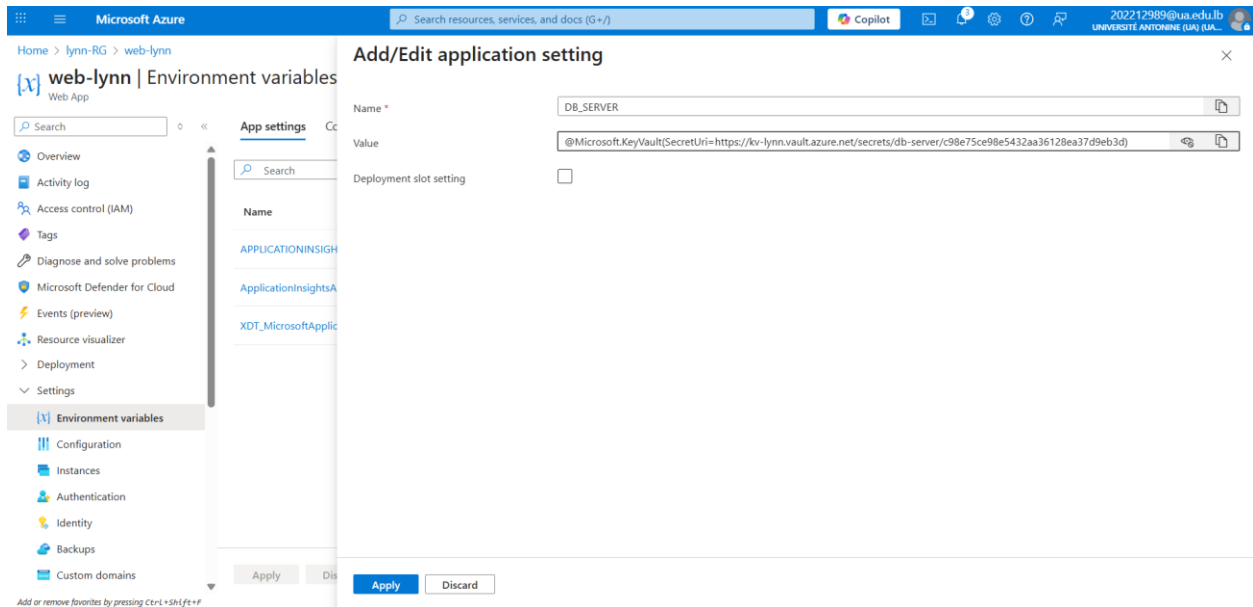


Figure 55_db-server

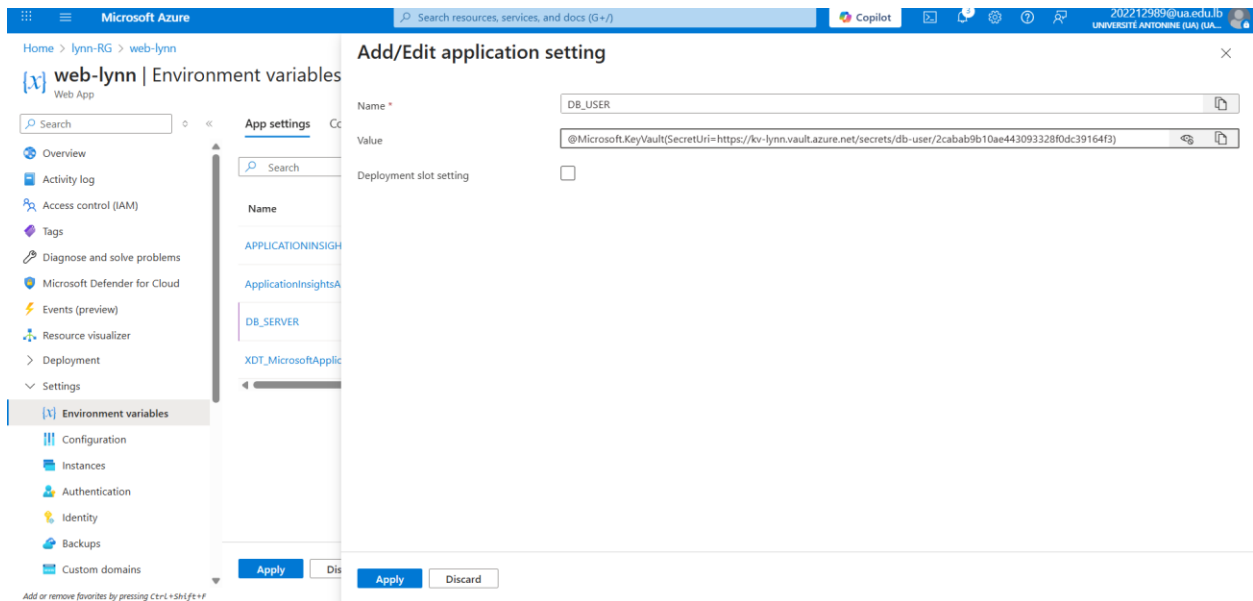


Figure 56_db-user

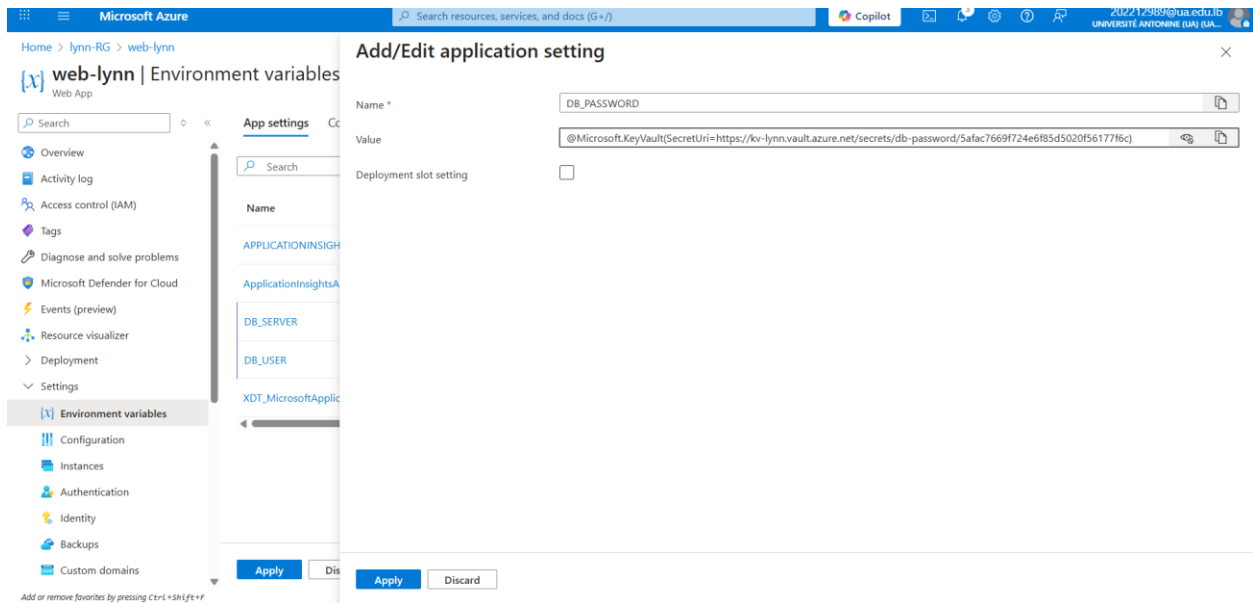


Figure 57_db-password

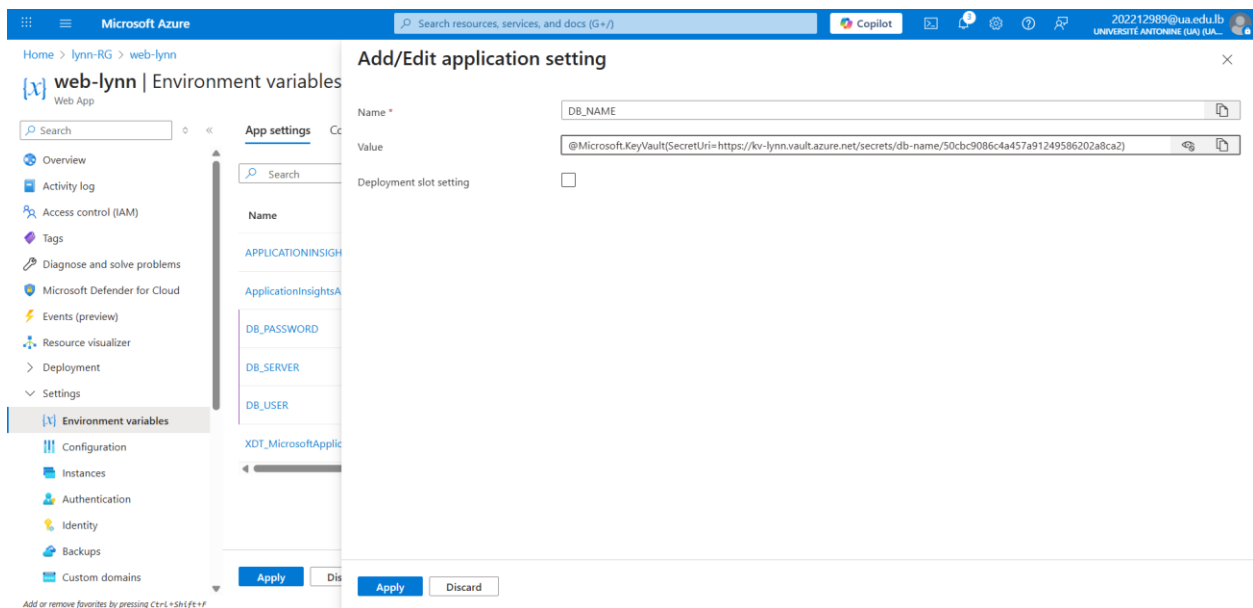


Figure 58_db-name

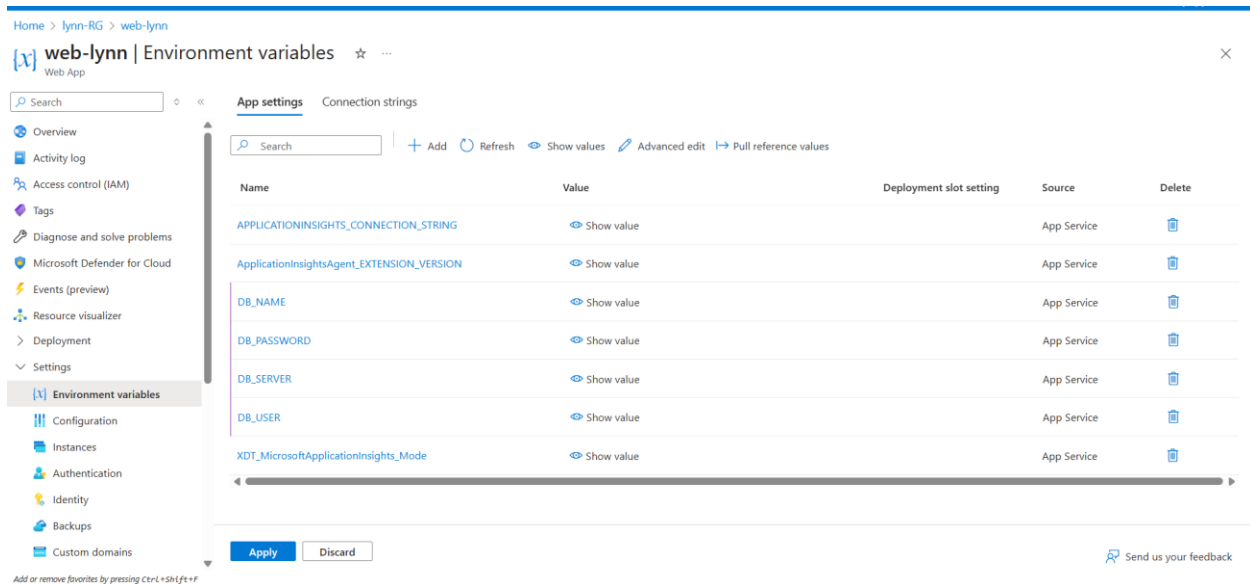


Figure 59_variables created

Save and restart web-lynn

Now my app will load secrets from Key Vault and connect to SQL.

6. Create Azure Function App

6.1 Create Function App Azure Portal

Search function app -> create -> consumption plan

[Home](#) > [Function App](#) > [Create Function App](#) >

Create Function App (Consumption) ...

all your resources.

Subscription * ⓘ

Resource Group * ⓘ [Create new](#)

Instance Details

Function App name * ✓
-cubgdxh4ftbeewgz.francecentral-01.azurewebsites.net

☒ Secure unique default hostname on. [More about this update](#) ⓘ

Operating System * ☒ Linux (legacy) ☐ Windows

i Flex Consumption is now the recommended serverless hosting plan for Azure Functions. [Learn more.](#)

Runtime stack *

Version *

Region *

[Review + create](#)

[< Previous](#)

[Next : Storage >](#)

Figure 60_basics tab

(I then changed the os to windows due to an error)

[Basics](#) [Storage](#) [Networking](#) [Monitoring](#) [Deployment](#) [Authentication](#) [Tags](#) [Review + create](#)

Storage

When creating a function app, you must create or link to a general-purpose Azure Storage account that supports Blobs, Queue, and Table storage. [Learn more](#) ⓘ

Storage account * ✓
[Create new](#)

i If you don't see a storage account, it may not be supported. [Storage account requirements](#)

Add an Azure Files connection



i Azure Files is used to enable certain features but you can create an app without one if you don't want to add another connection to your storage account. [What is Azure Files used for?](#)

Figure 61_storage tab

Home > Microsoft.Web-FunctionApp-Portal-ccde0c52-a722 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name : Microsoft.Web-FunctionApp-Portal-ccde0c52-a722 Start time : 23/11/2025, 18:33:05

Subscription : Lynn Haydar Correlation ID : 7ceae97e-efe9-4223-ab20-4edc258fe3ca

Resource group : lynn-RG

Deployment details

Resource	Type	Status	Operation details
func-lynn	Microsoft.Web/sites	OK	Operation details
ASP-lynnRG-9bc6	Microsoft.Web/serverfarms	OK	Operation details
func-lynn	Application Insights	OK	Operation details
func-lynn	Application Insights	OK	Operation details
storlynn	Storage account	OK	Operation details

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Find an Azure expert >

Figure 62_function app created

6.2. Enable Managed Identity

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

202212989@ua.edu.lb

UNIVERSITÉ ANTONINE (UA) (GAL...)

Home > Microsoft.Web-FunctionApp-Portal-ccde0c52-a722 | Overview > func-lynn

func-lynn | Identity

Function App

Search

Events (preview)

Resource visualizer

Functions

Deployment

Settings

Environment variables

Configuration (preview)

Instances

Durable Functions

Authentication

Azure OpenAI

Vector Database

Identity

Backups

Custom domains

Certificates

Networking

System assigned User assigned

A system assigned managed identity is restricted to one per resource and is tied to the lifecycle of this resource. You can grant permissions to the managed identity by using Azure role-based access control (Azure RBAC). The managed identity is authenticated with Microsoft Entra ID, so you don't have to store any credentials in code.

Save Discard Refresh Troubleshoot Got feedback?

Status

Off On

Add or remove favorites by pressing Ctrl+Shift+F

6.3. Give Function App access Key Vault

Got kv-lynn and steps like before

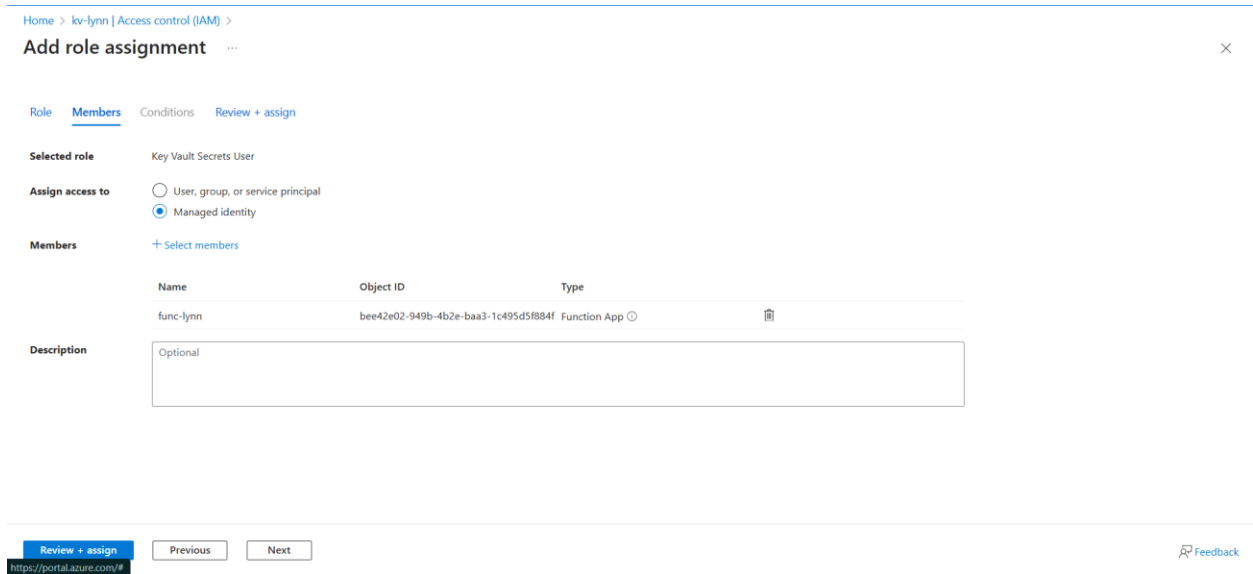


Figure 63_add role func app

now my functions can load SQL / configuration secrets.

7. Create Functions Locally

Inside CloudProject locally, create a new folder FunctionApp

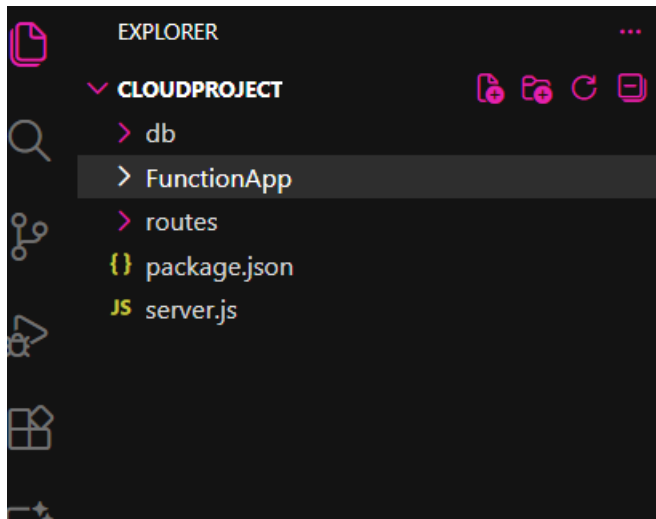


Figure 64_creating folder FunctionApp

7.1. Initialize the Function App

```
C:\Users\user\Desktop\uni\Cloud Computing\Project\CloudProject\FunctionApp>func init . --worker-runtime node --language javascript

The new Node.js programming model is generally available. Learn more at https://aka.ms/AzFuncNodeV4
Writing package.json
Writing .funcignore
Writing .gitignore
Writing host.json
Writing local.settings.json
Writing C:\Users\user\Desktop\uni\Cloud Computing\Project\CloudProject\FunctionApp\.vscode\extensions.json
Running 'npm install'...
```

Chat quota reached Prettier

Figure 65_initialize function app

7.2. Create the HTTP Trigger

```
module.exports = async function (context, req) {
    const quotes = [
        "Believe you can and you're halfway there.",
        "Small steps every day lead to big results.",
        "Stay positive, work hard, make it happen.",
        "Discipline is choosing what you want most.",
        "Every day is a chance to get better."
    ];

    // pick a random quote
    const random = quotes[Math.floor(Math.random() * quotes.length)];

    context.log("HttpFunction executed - quote returned");

    context.res = {
        status: 200,
        body: random
    };
};
```

Figure 66_http function

7.3. Create FetchAnalytics Function

```
const fetch = require("node-fetch");

module.exports = async function (context, req) {
  context.log("🔗 FetchAnalytics function triggered");

  try {
    const url = "https://web-lynn-avh0g7fxendagjbf.francecentral-01.azurewebsites.net/messages";

    context.log("📡 Fetching notes from:", url);

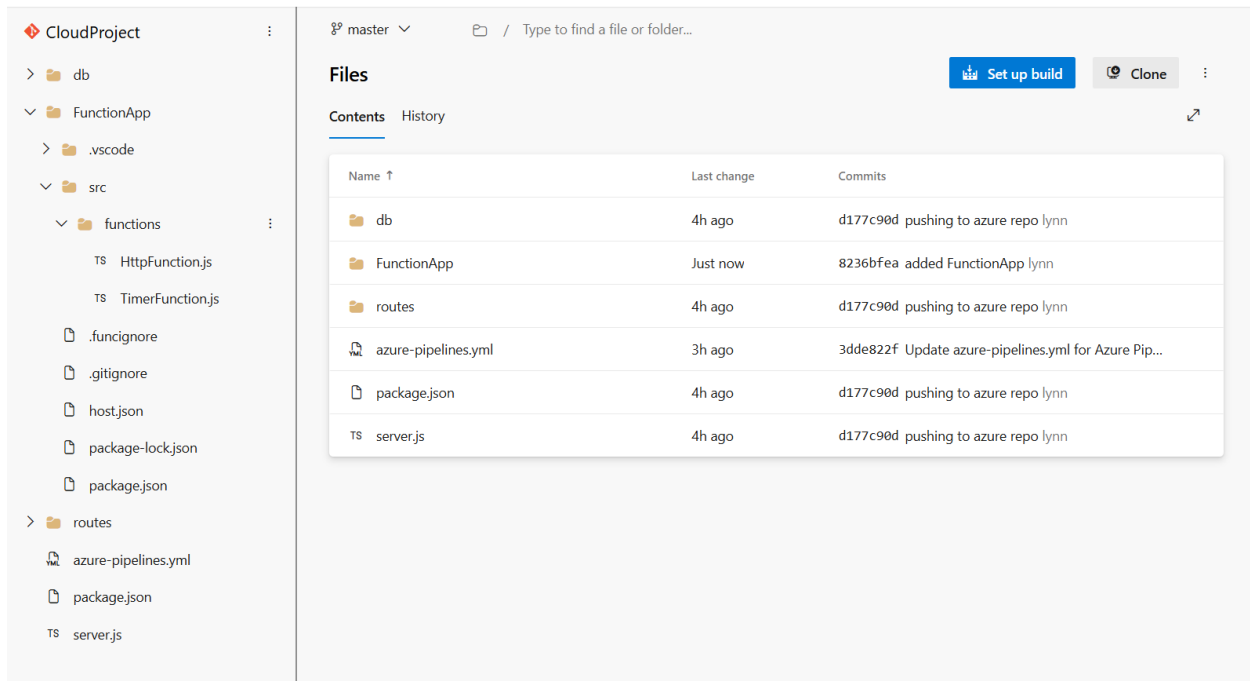
    const response = await fetch(url);
    const notes = await response.json();

    context.log("📄 Notes fetched: ${notes.length}");

    context.res = {
      status: 200,
      body: {
        message: "Analytics fetched successfully",
        totalNotes: notes.length,
        latestNote: notes[0] || null
      }
    };
  } catch (err) {
```

Figure 67_fetch analytics function

7.4. Push the changes to Azure Repos



The screenshot shows the Azure DevOps interface for a repository named 'CloudProject'. The 'Files' tab is active, displaying a table of files and their commit history. The left sidebar shows the file explorer with the 'functions' folder expanded.

Name ↑	Last change	Commits
db	4h ago	d177c90d pushing to azure repo lynn
FunctionApp	Just now	8236bfea added FunctionApp lynn
routes	4h ago	d177c90d pushing to azure repo lynn
azure-pipelines.yml	3h ago	3dde822f Update azure-pipelines.yml for Azure Pip...
package.json	4h ago	d177c90d pushing to azure repo lynn
server.js	4h ago	d177c90d pushing to azure repo lynn

Figure 68_pushed successfully

8. Create the CI pipeline for FunctionApp

This pipeline will:

- ✓ Install Node.js
- ✓ Install Azure Functions Core Tools
- ✓ Install dependencies for your FunctionApp
- ✓ Package it into a ZIP file
- ✓ Publish it as an artifact for the Release Pipeline

Create a new Pipeline

← CloudProject (11)

master

CloudProject / azure-pipelines-2.yml

```
1 trigger:
2   - main
3
4 pool:
5   name: 'Default'
6
7 steps:
8   Settings
9   - task: NodeTool@0
10    inputs:
11      versionSpec: '20.x'
12      displayName: 'Install Node.js 20'
13
14   # Install dependencies
15   - script: |
16       cd FunctionApp
17       npm install
18     displayName: "Install Function App Dependencies"
19
20   # Archive Function App (classic structure)
21   Settings
22   - task: ArchiveFiles@2
23     inputs:
24       rootFolderOrFile: '$(System.DefaultWorkingDirectory)/FunctionApp'
25       includeRootFolder: false
26       archiveType: 'zip'
27       archiveFile: '$(Build.ArtifactStagingDirectory)/functionapp.zip'
28       replaceExistingArchive: true
29       displayName: 'Archive Function App'
30
31   Settings
32   - task: PublishBuildArtifacts@1
33     inputs:
34       PathToPublish: '$(Build.ArtifactStagingDirectory)'
35       ArtifactName: 'functiondrop'
36       publishLocation: 'Container'
37       displayName: 'Publish artifact'
```

Figure 69_yaml pipeline functionapp

Save and Run

The screenshot displays the Azure DevOps interface for a build job. The top navigation bar shows the path: CloudProject / Pipelines / CloudProject (10) / 20251123.1. A search bar and user profile icon are on the right. The left sidebar shows the job list for 'Jobs in run #20251123.1' under 'CloudProject (10)'. The 'Job' is selected, showing a duration of 5m 49s. The job steps are listed below, all with green checkmarks indicating success:

- Initialize job (<1s)
- Checkout CloudProject... (7s)
- Install Node.js 20 (3s)
- Install Azure Funct... (1m 31s)
- Install Function Ap... (1m 14s)
- Archive Function A... (1m 22s)
- Publish Function A... (1m 28s)
- Post-job: Checkout Cl... (<1s)
- Finalize Job (<1s)
- Report build status (<1s)

The main panel shows the job details for 'Job'. It includes a green checkmark icon and the text 'Job'. Below this, the job parameters are listed:

- 1 Pool: Default
- 2 Agent: lynn-agent-vm
- 3 Started: Today at 8:53 PM
- 4 Duration: 5m 49s
- 5
- 6 Job preparation parameters

The job log shows the following steps:

```
41 [artifact] produced
```

Figure 70_build succeeded

The screenshot displays the 'Artifacts' section of the Azure DevOps interface. The 'Published' tab is selected, showing a table of artifacts:

Name	Size
functiondrop	526 MB
functionapp.zip	526 MB

Figure 71_artifact present

9. Create the Release Pipeline (CD) for the Function App

Releases -> new pipeline -> deploy function app template

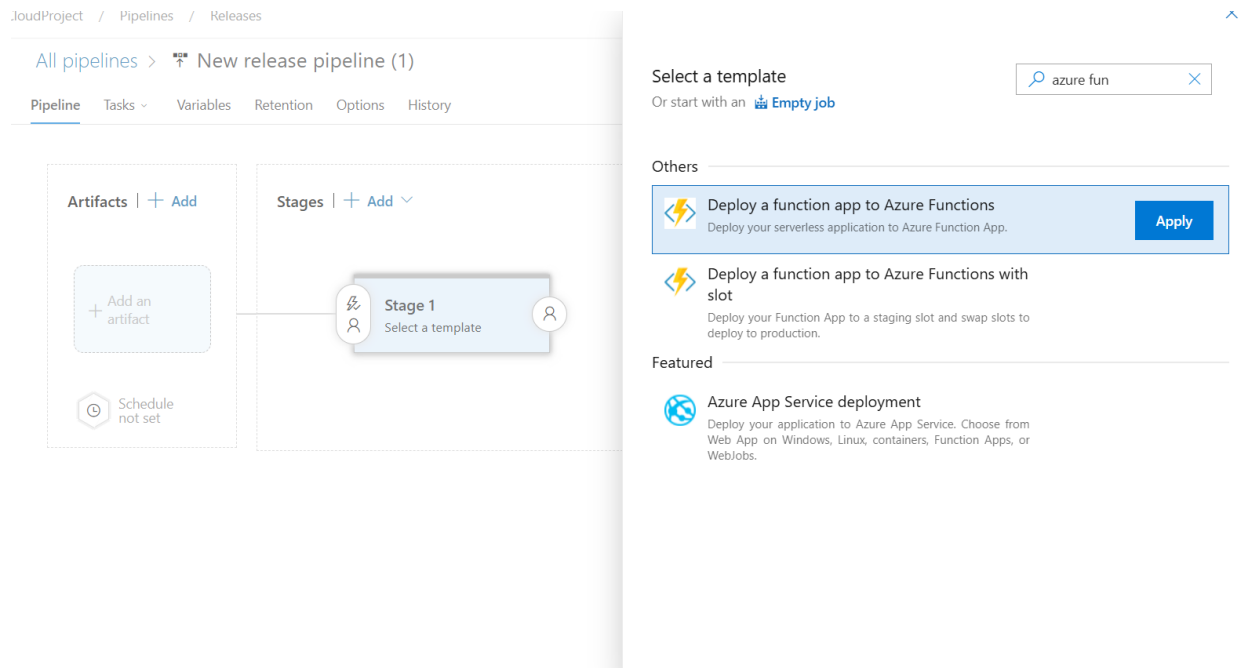


Figure 72_selecting template FunctionApp

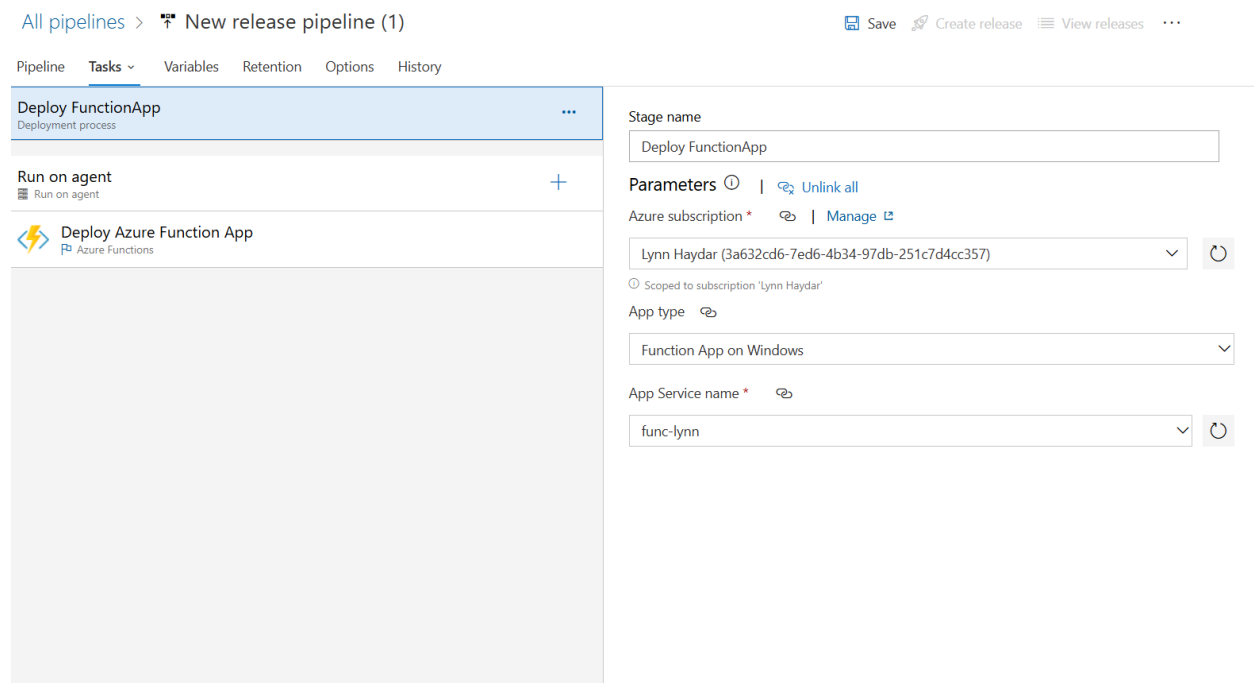


Figure 73_configure agent

Add the Artifact

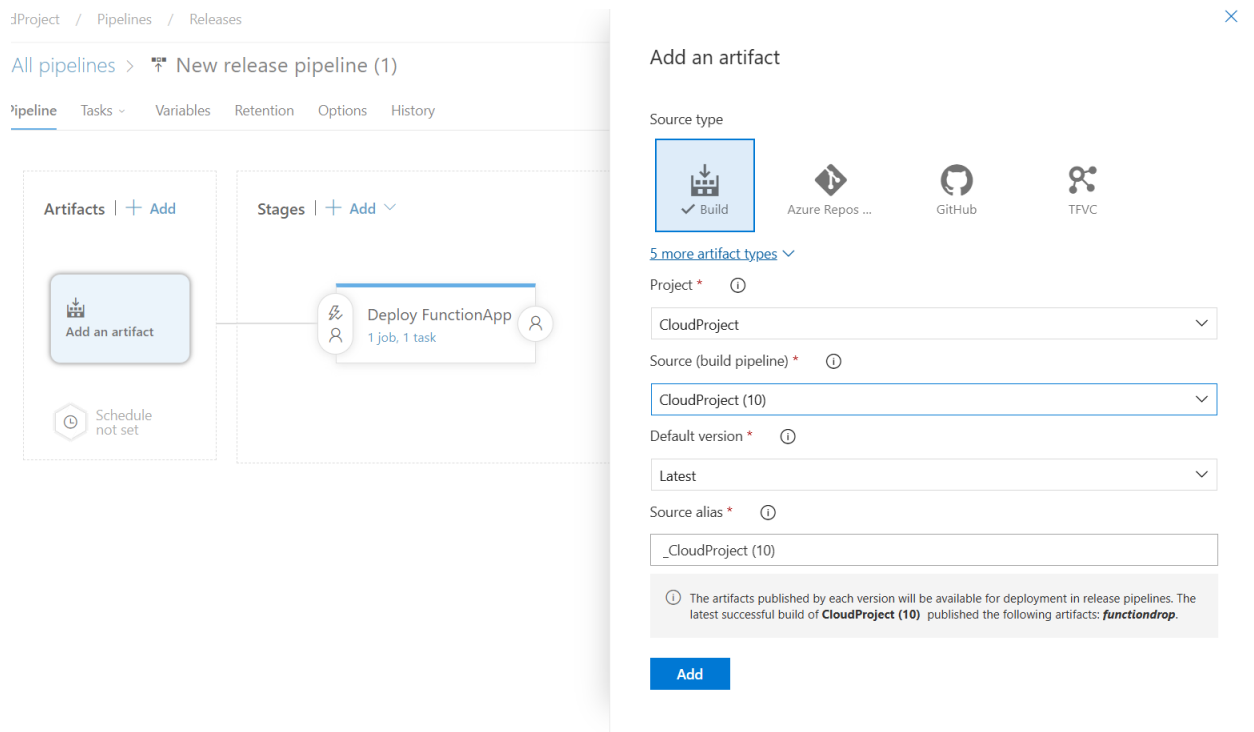


Figure 74_add artifact

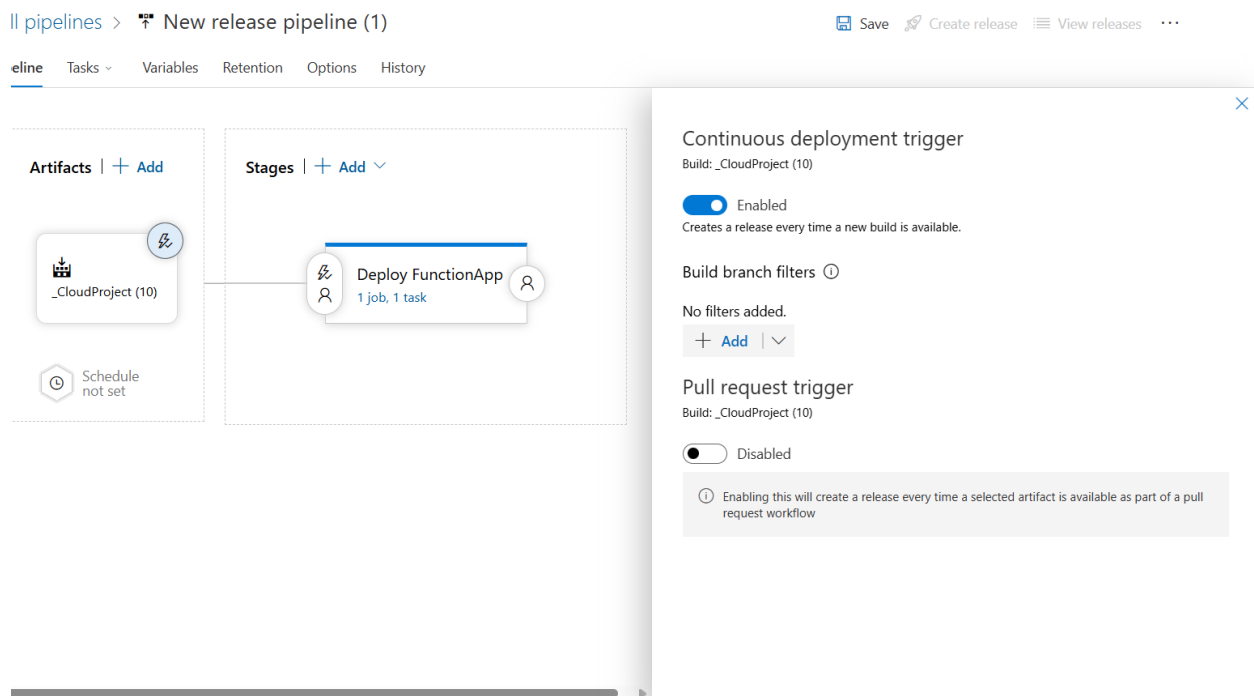


Figure 75_enable cd trigger

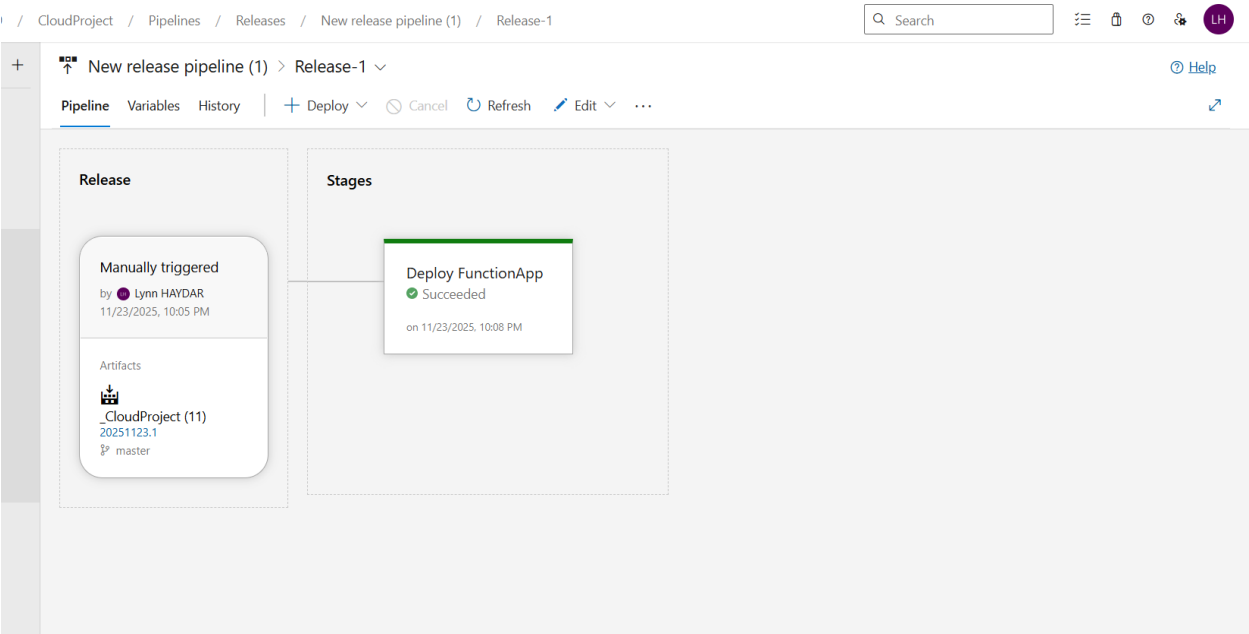


Figure 76_deployed successfully

Functions Properties Notifications (0)

() Set up local environment Refresh

Filter by name...

Name	Trigger	Status	Monitor
FetchAnalytics	HTTP	Enabled	Invocations and more
HttpFunction	HTTP	Enabled	Invocations and more

Figure 77_checking

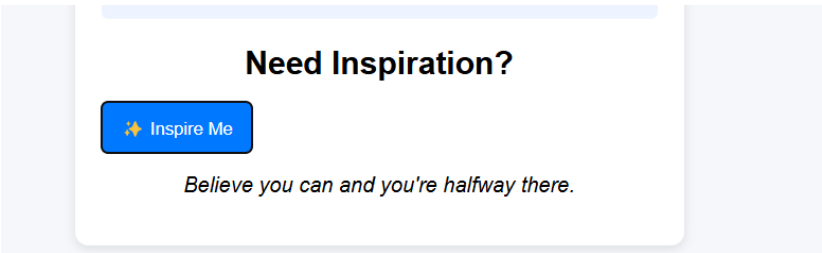


Figure 78_testing http

```
2025-12-02T19:15:16Z [Information] FetchAnalytics function triggered
2025-12-02T19:15:16Z [Information] Fetching notes from: https://web-lynn-avh0g7fxendagjbf.francecentral-01.azurewebsites.net/messages
2025-12-02T19:15:16Z [Information] Notes fetched: 5
2025-12-02T19:15:16Z [Information] Executed 'Functions.FetchAnalytics' (Succeeded, Id=c48e2e65-b302-4711-8f55-ab1d35b04ba7, Duration=59ms)
```

Figure 79_testing fetch analytics

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